

»» Chapter IV ««

COMPLEX DIFFERENTIAL ANALYSIS

PREFATORY REMARKS

We are now ready to undertake a study of the differential calculus in the complex domain. Some of the initial parts of the program are little more than paraphrase and transplantation of the one real variable theory. However, to reveal our pedagogic hand, the time devoted to review, extension, and comparison is a useful period of synthesis for many students. Often some of the most basic ideas of real differential analysis become alive for them at just this period of development. After these preliminaries we shall continue by studying the differentiability of functions represented by power series (§3) and shall find that we have at our disposal a method for obtaining solutions of a number of important differential equations—a method whose power the mathematicians of the eighteenth century found intoxicating: try to find a solution admitting a power series representation (method of undetermined coefficients for power series).

The way is opened for the study of that function the worker in complex function theory would most reluctantly surrender, the *exponential function* (on $\mathbf{C}$). The moment is at hand to prepare a number of fundamental working tools: the polar representation of complex numbers, logarithms, arguments, powers. The apparatus so introduced allows us to treat early the important Poisson integral representation of the real part of an analytic function (§5). In the present chapter we take for granted familiarity with the elementary

properties of the Riemann integral. However, our wish to keep the exposition essentially “self-contained” will not be violated for the Riemann–Stieltjes integral will be treated in Ch. V. Thus we shall account for the material concerning the Riemann integral used in the present chapter.

An important question is the relation between differential analysis involving one complex variable and differential analysis involving two real variables (§7), in other words, the link between the differential analysis of f , a function of a complex variable, and the function of two real variables obtained from f with the mediation of the canonical map σ of Ch. II. We arrive at the Cauchy–Riemann equation(s) (§7) and the connection with harmonic functions (§8). To pave the way for §§7 and 8 it will be sound pedagogically to review some aspects of two real variable theory. We shall do so in §6.

The Gauss theorem on the zeros of the derivative of a polynomial is treated in §9. Its statement fits nicely in the realm of one complex variable differential analysis. Its proof brings into play the notion of the logarithmic derivative, the multiplicative decomposition of a polynomial, and the attractive notion of convexity.

1. COMPLEX-VALUED FUNCTIONS OF ONE REAL VARIABLE

We consider $f: V \rightarrow \mathbb{C}$ where V is a neighborhood (in the sense of the $\mathbb{R}$ -topology) of $a \in \mathbb{R}$, and recall that f is said to *possess a derivative at a* provided that $[f(x) - f(a)]/(x - a)$ has a limit at a (of course, it is to be emphasized, in $\mathbb{C}$). The notation “ $[f(x) - f(a)]/(x - a)$ ” intends a function with domain $V - \{a\}$. When f possesses a derivative at a , we term

$$\lim_a \frac{f(x) - f(a)}{x - a}$$

the *derivative of f at a* and denote it by $f'(a)$. We say “ $f'(a)$ exists” for “ f possesses a derivative at a .” When we assert “ $f'(a) = A$ ” where $A \in \mathbb{C}$, we mean that f possesses a derivative at a and the derivative of f at a is A . The notion of derivative may be given an elegant and useful equivalent definition due to Carathéodory: f possesses a derivative at a provided that there exists $g: V \rightarrow \mathbb{C}$, continuous at a , such that

$$f(x) - f(a) = g(x)(x - a), \quad x \in V. \quad (1.1)$$

EXERCISES

1.1. If f has a derivative at a , then f is continuous at a .

1.2. Show that there is at most one g satisfying the imposed conditions. When $f'(a)$ exists, $f'(a) = g(a)$.

1.3. $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$ both possess derivatives at a if and only if f does. In this case $f'(a) = u'(a) + i v'(a)$. [Here “ $\operatorname{Re} f$ ” for “ $\operatorname{Re} \circ f$ ” and “ $\operatorname{Im} f$ ” for “ $\operatorname{Im} \circ f$.”]

How incisive Carathéodory's formulation is may be seen by applying it to the proof of the *chain rule* for the derivative of a composite function: *Given $f: V \rightarrow U$, $g: U \rightarrow \mathbf{C}$, where V is a neighborhood of $a \in \mathbf{R}$, U is a neighborhood of $b = f(a) \in \mathbf{R}$, both V and U being neighborhoods in the sense of the $\mathbf{R}$ -topology, $f'(a)$ and $g'(b)$ exist. Let $h = g \circ f$. Then $h'(a) = g'(b)f'(a)$.*

Here is a proof using the formulation of Carathéodory. We have

$$f(x) - b = \varphi(x)(x - a), \quad x \in V,$$

where $\varphi: V \rightarrow \mathbf{R}$ and is continuous at a , and

$$g(x) - g(b) = \psi(x)(x - b), \quad x \in U,$$

where $\psi: U \rightarrow \mathbf{C}$ and is continuous at b . Hence

$$\begin{aligned} h(x) - h(a) &= \psi[f(x)][f(x) - b] \\ &= [\psi \circ f](x) \varphi(x)(x - a), \quad x \in V. \end{aligned}$$

The proof is completed by noting that $(\psi \circ f)\varphi$ is continuous at a and that its value at a is $g'(b)f'(a)$.

When V is not assumed to be a neighborhood of a but is assumed to contain some segment $[a, b]$, $a < b$, we may consider the existence of a limit at a for $[f_+(x) - f_+(a)]/(x - a)$, where $f_+ = f|_{(V \cap [a, +\infty[)}$. When it exists, it is termed the *right-handed derivative of f at a* and is denoted $f'_+(a)$. Analogous remarks hold for the left-handed derivative, the notation $f'_-(a)$ now being used.

Sign of derivative. We note that if $F: X \rightarrow \mathbf{R}$, where X is a topological space, and F is continuous at $a \in X$ and $F(a) > 0$, then there exists a neighborhood V of a such that $F(x) > 0$, $x \in V$. Next we introduce the *sign function*, $\operatorname{sg}: \mathbf{C} \rightarrow \mathbf{C}$, defined by

$$\operatorname{sg} 0 = 0; \quad \operatorname{sg} z = \frac{z}{|z|}, \quad z \neq 0, \quad (1.2)$$

and note that sg is multiplicative: $\operatorname{sg}(zw) = (\operatorname{sg} z)(\operatorname{sg} w)$. Let us now turn to f possessing a derivative at a , f real valued, $f'(a) \neq 0$. From (1.1) we have

$$\operatorname{sg}[f(x) - f(a)] = \operatorname{sg}[f'(a)] \operatorname{sg}(x - a) \quad (1.3)$$

in some neighborhood V_1 of a . This is the starting point of important considerations of the elementary differential calculus. From (1.3) follows: if

$f(a) \geq f(x)$ for x in some neighborhood of a , that is, if f has a relative maximum at a , or if f has a relative minimum at a , then $f'(a) = 0$.

We now see that the theorem of Rolle and the mean-value theorem of the differential calculus are easily concluded thanks to the observation just made and the theorem of Weierstrass concerning the existence of a maximum and a minimum of a continuous real-valued function. The mean-value theorem states: Let $f: [a, b] \rightarrow \mathbf{R}$, $-\infty < a < b < +\infty$. Suppose that f is continuous on $[a, b]$ and that $f'(x)$ exists for all $x \in]a, b[$. Then there exists $c \in]a, b[$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

EXERCISES

1.4. Treat the special case where $f(a) = f(b)$ (theorem of Rolle) by showing the existence of $c \in]a, b[$ at which f attains either its maximum or its minimum.

1.5. Establish the mean-value theorem by applying the theorem of Ex. 1.4 to g defined by

$$g(x) = f(x) - \left[f(a) \frac{b-x}{b-a} + f(b) \frac{x-a}{b-a} \right], \quad a \leq x \leq b.$$

1.6. Show that if the hypotheses of the mean-value theorem are fulfilled and $f'(x) > 0$ for all $x \in]a, b[$, then f is increasing on $[a, b]$, that is, $f(x) < f(y)$ when $a \leq x < y \leq b$. Formulate and establish corresponding theorems when $f'(x) < 0$ (resp., $f'(x) \geq 0$, $f'(x) \leq 0$) for all $x \in]a, b[$.

Special mention should be made of the case where $f'(x) = 0$ for all $x \in]a, b[$. Here f is constant, that is, $f(x) = f(a)$, $a \leq x \leq b$. Frequent use will be made of this fact which is valid also when $f: [a, b] \rightarrow \mathbf{C}$. Prove.

1.7. The example of this exercise shows that one must not expect too much from the fact that the derivative at a given point is not zero. It is fallacious to think that if $f'(a) > 0$, then f is increasing (or even never decreasing) on some $]a - h, a + h[$, $h > 0$. Let $a = 0$; let $\alpha \in \mathbf{R}$, $\alpha > 1$; let $f(0) = 0$, $f(x) = x + |x|^\alpha \sin(x^{-1})$ for $x \neq 0$. Show that $f'(0) = 1$, but that when $1 < \alpha \leq 2$ the function f is not never decreasing on any interval $] -h, h[$, $h > 0$. Investigate the case where $\alpha > 2$.

Primitive. Given $f: I \rightarrow \mathbf{C}$, where I is a connected subset of $\mathbf{R}$ containing more than one point, we are led to ask whether there exists $g: I \rightarrow \mathbf{C}$ such that $g'(x) = f(x)$, $x \in I$, one-sided derivatives replacing the derivative at $\min I$, $\max I$ when I has one or both. Such a g is termed a *primitive* of f . In general, the study of the existence of a primitive is difficult in real analysis—we shall see that the facts are much simpler in complex analysis—but it is easy to say something about the totality of primitives, for if g and h are both primitives of f , then $(h - g)'(x) = 0$, $x \in I$, and, consequently, $h = g + k$, where k is a

complex constant. On the other hand, $g + k$ is a primitive of f for every complex constant k . Thus, if f possesses a primitive g , the totality of primitives of f is $\{g + k : k \text{ constant}\}$.

Remarks on the calculus of derivatives. In the elements of analysis we are led at once to consider the existence of the derivative (and its value) for the usual “combinations” of functions: sum, product, quotient, composition. The first three are so familiar and so much part of the mathematical nursery that we pass over them without comment. The reader may find it amusing to test the Carathéodory approach on them. The case of composition was treated because it merited attention. All this will allow itself to be transplanted to the complex domain without any effort. A more serious matter is the question of the derivative of the *inverse function*, which we shall treat using once again Carathéodory’s formulation of the derivative. In the simple real case we find the prototype of a very general argument that may be employed in the study of the inverse of a “well-behaved” map from a region of euclidean space into euclidean space. We show

Theorem 1.1: *Let $f:]a, b[\rightarrow \mathbf{R}$, $a < b$. Let $a < c < b$. Suppose that f is univalent and continuous on $]a, b[$ and that $f'(c)$ exists and $f'(c) \neq 0$. Let $d = f(c)$, let g denote the inverse of f . Then $g'(d) = [f'(c)]^{-1}$.*

PROOF: The domain of g is of the form $]A, B[$, $A < B$. Also $A < d < B$ and g is continuous (cf. Ex. 3.4, Ch. III). Using the Carathéodory formulation we have

$$f(x) - d = \varphi(x)(x - c), \quad a < x < b,$$

where φ is continuous on $]a, b[$ and further $\varphi(x) \neq 0$, $a < x < b$. Hence

$$y - d = f[g(y)] - d = \varphi[g(y)][g(y) - c], \quad A < y < B,$$

so that

$$g(y) - c = \frac{1}{\varphi[g(y)]}(y - d), \quad A < y < B.$$

Now $1/(\varphi \circ g)$ is continuous on $]A, B[$. The theorem follows.

EXERCISE

1.8. Formulate and prove a corresponding theorem when the domain of f includes an endpoint $a \in \mathbf{R}$ such that f has a one-sided nonzero derivative at a .

2. COMPLEX-VALUED FUNCTIONS OF ONE COMPLEX VARIABLE

The material of the first two paragraphs of §1, this chapter, goes over to the complex case when the following changes are made: a, b are now to be taken in $\mathbb{C}$, and V and U are to be neighborhoods of a and b , respectively, in the sense of the topology of $\mathbb{C}$. The same goes for the usual rules for the derivative of sum, product, and quotient. We see that a polynomial possesses a derivative at each point of $\mathbb{C}$. We shall want the chain rule when $a \in \mathbb{R}$ and V is a neighborhood of a in the sense of the $\mathbb{R}$ -topology, while $b \in \mathbb{C}$. It holds, of course.

Suppose that $f: A \rightarrow \mathbb{C}$ where A is a region in $\mathbb{C}$. If f is constant, then $f'(z) = 0, z \in A$. We now show that the converse holds: if $f'(z) = 0, z \in A$, then f is constant.

PROOF: Let $\Delta(a; r) \subset A$. For $z \in \Delta(a; r)$ we introduce

$$g(t) = f[a(1 - t) + zt], \quad 0 \leq t \leq 1,$$

and conclude by the chain rule $g'(t) = 0, 0 \leq t \leq 1$. Hence

$$f(z) = g(1) = g(0) = f(a).$$

We now fix $a \in A$. Let $O = \{f(z) = f(a)\}$. From what we have just seen O is open. By the continuity of f , $A - O = \{f(z) \neq f(a)\}$ is open. Since A is connected and $a \in O$, $O = A$ and the theorem follows.

We define primitive as in the real theory. Given $f: O \rightarrow \mathbb{C}$, O open in $\mathbb{C}$, g is a *primitive* of f provided that g has domain O and $g' = f$. If O is a region and there exists a primitive, g , of f , then the totality of primitives of f is given by $g + c$, c constant.

Terminology. In complex function theory we are interested in functions that possess a derivative at each point of some neighborhood of a given point. It is desirable to have suitable terminology. Let $a \in \mathbb{C}$, let $f: V \rightarrow \hat{\mathbb{C}}$ where V is a neighborhood of a in the sense of the topology of $\hat{\mathbb{C}}$. We say that f is *analytic* at a provided that, for some open $U, a \in U \subset \mathbb{C} \cap V, f|U$ takes values in $\mathbb{C}$ and possesses a derivative at each point of U . Given an open set $O \subset \mathbb{C}$ and $f: A \rightarrow \hat{\mathbb{C}}, O \subset A \subset \hat{\mathbb{C}}$, we say that f is *analytic in O* provided that f is analytic at each point of O . Given f analytic in O , we say that f is *analytic on O* provided that O is the domain of f .

EXERCISE

2.1. Given $f: A \rightarrow \hat{\mathbb{C}}$, where $A \subset \hat{\mathbb{C}}$. Let E denote the set of $z \in \mathbb{C}$ such that f is analytic at z . Then E is open.

3. POWER SERIES

It will be convenient to use the notation $\Delta(a; +\infty)$, $a \in \mathbb{C}$. It is to have the expected meaning $\{|z - a| < +\infty\} = \mathbb{C}$. A venerable fact of the theory of power series is

Theorem 3.1: Given $\sum_0^\infty a_k z^k$ with radius of convergence ρ , $0 < \rho \leq +\infty$. Let f denote the sum on $\Delta(0; \rho)$. Then f is analytic on $\Delta(0; \rho)$ and

$$f'(z) = \sum_0^\infty (k+1)a_{k+1}z^k, \quad |z| < \rho. \quad (3.1)$$

PROOF: The following proof is a slight variant of an elegant one given by v. Stachó [12, vol. I, p. 38]. We first show that $\sum_0^\infty (k+1)a_{k+1}z^k$ has radius of convergence ρ . Consequently the right-hand side of (3.1) is meaningful. Indeed, given that $0 < |w| < \rho$, there exists a positive real M such that

$$\begin{aligned} |(k+1)a_{k+1}z^k| &= |a_{k+1}w^{k+1}| \cdot (k+1) \left| \frac{z}{w} \right|^k \cdot \frac{1}{|w|} \\ &\leq M(k+1) \left| \frac{z}{w} \right|^k, \end{aligned}$$

$|z| < |w|$, $k \in \mathbb{N}$. Convergence of $\sum_0^\infty (k+1)a_{k+1}z^k$, $|z| < \rho$, follows on using the convergence of $\sum_0^\infty (k+1)x^k$, $|x| < 1$. The radius of convergence of $\sum_0^\infty (k+1)a_{k+1}z^k$, say ρ' , is at least ρ . Since

$$|a_{k+1}z^{k+1}| \leq |(k+1)a_{k+1}z^k| |z|,$$

we see that $\rho \geq \rho'$. Hence $\rho = \rho'$.

We fix $b \in \Delta(0; \rho)$. Let $|b| < r < \rho$. For k a positive integer we introduce φ_k by

$$\varphi_k(z) = \frac{z^k - b^k}{z - b}, \quad z \neq b; \quad \varphi_k(b) = kb^{k-1}.$$

Actually, φ_k is a polynomial of degree $k - 1$ and

$$\begin{aligned} |\varphi_k(z)| &= \left| \sum_{j=0}^{k-1} z^{k-1-j} b^j \right| \\ &\leq \sum_{j=0}^{k-1} |z|^{k-1-j} |b|^j \\ &\leq k r^{k-1}, \quad |z| \leq r. \end{aligned}$$

Now

$$f(z) - f(b) = (z - b) \sum_{k=1}^{\infty} a_{k+1} \varphi_{k+1}(z), \quad |z| < \rho.$$

Since $\sum_{k=0}^{\infty} (k+1) |a_{k+1}| r^k$ converges, we conclude by the Weierstrass M -test that $\sum_{k=0}^{\infty} a_{k+1} \varphi_{k+1}$ converges uniformly in $\Delta(0; \rho)$ and hence that its sum on $\Delta(0; \rho)$ is continuous. Consequently

$$f'(b) = \sum_{k=0}^{\infty} a_{k+1} \varphi_{k+1}(b).$$

The theorem follows.

EXERCISES

3.1. Give a second proof of Theorem 3.1 starting from the observation that

$$\begin{aligned} \left| \frac{z^{k+1} - b^{k+1}}{z - b} - (k+1)b^k \right| &\leq \frac{(|b| + |z - b|)^{k+1} - |b|^{k+1}}{|z - b|} - (k+1)|b|^k \\ &\leq (k+1)[(|b| + |z - b|)^k - |b|^k] \\ &\leq (k+1)k(|b| + |z - b|)^{k-1} |z - b|, \end{aligned}$$

$z \neq b, k \in \mathbf{N}$.

3.2. Starting with the power series of Theorem 3.1 show that $\sum_1^{\infty} (a_{k-1}/k)z^k$ has radius of convergence ρ and that g , its sum on $\Delta(0; \rho)$, is the primitive of f which takes the value 0 at 0. What notational gloss is appropriate here or, in other words, what is the zeroth partial sum?

3.3. Given $\sum_0^{\infty} a_k z^k$ with radius of convergence > 0 . Suppose that $a_0 = 0$ but that $a_k \neq 0$ for some k . Show that there exists ε positive such that for each $u \in C(0; 1)$ the function $|\sum_0^{\infty} a_k (ru)^k|^2, r \in [0, \varepsilon]$, is increasing.

3.4. Let g be the primitive of the function $(1+z)^{-1}$ with domain $\Delta(0; 1)$ satisfying $g(0) = 0$. Show that

$$\lim_1 g = \sum_{k=0}^{\infty} (-1)^k (k+1)^{-1}.$$

Show that g is not representable as the quotient of polynomials. Let h denote the primitive of the function $(1 + z^2)$ with domain $\Delta(0; 1)$ satisfying $h(0) = 0$. Then

$$\lim_1 h = \sum_0^{\infty} (-1)^k (2k + 1)^{-1}.$$

Interpret the limit values in terms of the elementary functions encountered in the calculus.

3.5. Show that the convergence set of $\sum_0^{\infty} (k + 1)a_{k+1}z^k$ is contained in the convergence set of $\sum_0^{\infty} a_k z^k$.

Taylor series. We recall the following definition from elementary analysis. Let $f: V \rightarrow \mathbb{C}$ where V is a neighborhood (in the sense of the topology of $\mathbb{R}$) of a . It is supposed that for some interval $I =]a - h, a + h[$, h positive, the function $f|I$ has derivatives on I of all orders ($f|I$ is then said to be of class C^{∞}). By the Taylor series of f associated with a is meant

$$\sum_0^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

Here $f^{(k)}(a)$ is the k th derivative of f at a , $f^{(0)}$ being f .

The notion of a Taylor series makes sense in the complex domain. Let us suppose that $\sum_0^{\infty} a_k z^k$ is a power series with radius of convergence $\rho > 0$ and let f denote its sum on $\Delta(0; \rho)$. We see by repeated differentiation that

$$f^{(n)}(z) = \sum_0^{\infty} b_k^{(n)} z^k, \quad |z| < \rho,$$

where

$$b_k^{(n)} = a_{n+k} \left[\prod_0^{n-1} (n + k - j) \right], \quad k, n \in \mathbb{N},$$

and $f^{(n)}$ is defined recursively by $f^{(0)} = f$, $f^{(n+1)}(z) = f^{(n)'}(z)$, $|z| < \rho$. Hence taking $k = 0$ in $b_k^{(n)}$ we conclude

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

We obtain the basic conclusion: If $\sum_0^{\infty} a_k z^k$ has radius of convergence $\rho > 0$ and f is its sum on $\Delta(0; \rho)$, then $\sum_0^{\infty} a_k z^k$ is the Taylor series of f associated with 0.

We may now answer the question: Let f be as in the definition given above of a Taylor series. When does the Taylor series of f associated with a have a sum

on some $(\mathbf{R}-)$ neighborhood which agrees with f ? Indeed, the power series

$$\sum_0^{\infty} \frac{f^{(k)}(a)}{k!} z^k$$

must in this case have radius of convergence $\rho > 0$. Let g denote its sum on $\Delta(0; \rho)$. We have $g(x - a) = f(x)$, $x \in]a - h, a + h[$ for some positive h . Hence a *necessary* condition for f to fulfill the imposed requirement is that there exist a power series with radius of convergence $\rho > 0$ such that its sum g on $\Delta(0; \rho)$ satisfies $f(x) = g(x - a)$ for $x \in]a - h, a + h[$ for some positive h . The *sufficiency* is immediately verified by successive differentiations: $f^{(k)}(a) = g^{(k)}(0)$, $k \in \mathbf{N}$.

EXERCISES

3.6. Given $\sum_0^{\infty} a_k z^k$ has radius of convergence $\rho > 0$ and f is its sum on $\Delta(0; \rho)$. Show that if $\sum_0^{\infty} f^{(k)}(0)$ converges, then $\rho = +\infty$ [98].

3.7. Given $b \in \mathbf{C}$ and $(a_k): \mathbf{N} \rightarrow \mathbf{C}$. Show that the convergence set of $\sum_0^{\infty} a_k(z - b)^k$ is the image under the translation taking z into $z + b$ of the convergence set of $\sum_0^{\infty} a_k z^k$ and that if the radius of convergence of the latter series is $\rho > 0$ and g is the sum of $\sum_0^{\infty} a_k(z - b)^k$ on $\Delta(b; \rho)$, then $g^{(k)}(b) = k! a_k$, $k \in \mathbf{N}$.

The exponential function in the complex domain. There are many approaches. If we seek a function that is the sum of a power series with radius of convergence $+\infty$ having $\exp$, as defined in §5, Ch. I, for its restriction to $\mathbf{R}$, then at least we have introduced unicity into the problem. There is at most one such function by the identity theorem for power series. If we wished to attack the question by examining the Taylor series of $\exp$ associated with 0, we could reach a satisfactory conclusion. Here we shall attack the problem via a differential equation. We seek, to put it roughly, a power series solution of the differential equation

$$y' = y, \quad (3.2)$$

satisfying the initial condition $y(0) = 1$.† Let us explore the question. Suppose that $\sum_0^{\infty} a_k z^k$ has radius of convergence $\rho > 0$ and that f , its sum on $\Delta(0; \rho)$, satisfies $f(0) = 1$ and $f'(z) = f(z)$ when $|z| < r$, for some $r \in]0, \rho]$ (and hence, *a posteriori*, when $|z| < \rho$). Using the condition $f(0) = 1$, Theorem 3.1 of the present section, and the uniqueness theorem for power series we obtain

$$a_0 = 1, \quad (k + 1)a_{k+1} = a_k, \quad k \in \mathbf{N}. \quad (3.3)$$

† The function $\exp$ satisfies (3.2), the terms involved being taken in the sense of $\mathbf{R}$, and $\exp(0) = 1$.

Hence there is at *most one* such power series, for necessarily $a_k = (k!)^{-1}$ if there is a solution. The uniquely available candidate fulfills the imposed conditions. First, $\sum_0^\infty (k!)^{-1} z^k$ has radius of convergence $+\infty$. Let E denote its sum on $\mathbf{C}$. Since $((k!)^{-1})$ is the solution of the system (3.3), we see that $E(0) = 1$ and that $E' = E$. We agree to term E the *exponential function* (on $\mathbf{C}$) and proceed to show that

$$E|_{\mathbf{R}} = \exp. \quad (3.4)$$

Once this has been accomplished we shall not hesitate to denote E by $\exp$ and to speak of it simply as *the exponential function*. We shall denote $E(z)$ by both the traditional $\exp(z)$ and the ancient e^z . A justification for the latter notation is to be found in the addition theorem and the conventions made on powers (cf. the discussion at the end of §4, this chapter). To avoid difficulties we use e^z scrupulously as $E(z)$; in case of need, one should furnish a gloss.

We first show that E satisfies the *addition theorem*:

$$E(a + b) = E(a) E(b), \quad a, b \in \mathbf{C}. \quad (3.5)$$

The following elegant proof based on the differential equation (3.2) was given essentially by Carathéodory [21, vol. I, p. 233]. Let $f(z) = E(z) E(a + b - z)$, $z \in \mathbf{C}$. Then $f'(z) = 0$, $z \in \mathbf{C}$. Hence $f(a) = f(0)$ and (3.5) follows. Taking $b = -a$ in (3.5), we see that $E(a) \neq 0$. We conclude that $E(a) > 0$ for $a \in \mathbf{R}$.

It is elementary to see by the binomial theorem applied to

$$\left(1 + \frac{1}{n+1}\right)^{n+1}$$

that

$$\sum_0^m (k!)^{-1} \leq \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n+1}\right)^{n+1} \leq E(1), \quad m \in \mathbf{N},$$

whence $E(1) = e$. Since, in addition, $E|_{\mathbf{R}}$ satisfies the addition theorem and is continuous, $E|_{\mathbf{R}} = \exp$.

We introduce λ , the primitive of $(1+z)^{-1}$ with domain $\Delta(0; 1)$, satisfying $\lambda(0) = 0$. We consider $f = \exp \circ \lambda$ and note that

$$f'(z) = \frac{1}{1+z} f(z), \quad |z| < 1,$$

and hence $f(z) (1+z)^{-1}$ has derivative equal to 0 at each point of $\Delta(0; 1)$, so that $f(z) (1+z)^{-1} = f(0) = 1$. Hence the useful identity

$$\exp[\lambda(z)] = 1 + z, \quad |z| < 1. \quad (3.6)$$

By the addition theorem for the exponential function,

$$\exp[n \lambda(z)] = (1 + z)^n, \quad |z| < 1,$$

for $n \in \mathbf{N}$. Taking $z = (n + 2)^{-1}$ and replacing n by $n + 2$, we obtain a second proof that $\exp(1) = e$ on considering the limit as n tends to ∞ . It is to be noted that $\lim_0 z^{-1} \lambda(z) = \lambda'(0) = 1$.

EXERCISES

3.8. Establish the addition theorem for the exponential function using the Cauchy multiplication rule for power series.

3.9. Abel's binomial series. Let $\alpha \in \mathbf{C}$. Consider the differential equation

$$(1 + z)y' = \alpha y. \quad (3.7)$$

Seek power series with radius of convergence $\rho > 0$ having sum S_α on $\Delta(0; \rho)$ satisfying $S_\alpha(0) = 1$ and (3.7). Show that there is exactly one such series, that $\rho = +\infty$ when $\alpha \in \mathbf{N}$ and $\rho = 1$ when $\alpha \in \mathbf{C} - \mathbf{N}$, and that the k th coefficient is $\binom{\alpha}{k}$ defined recursively by

$$\binom{\alpha}{0} = 1, \quad \binom{\alpha}{k+1} = \binom{\alpha}{k} \frac{\alpha - k}{k+1}, \quad k \in \mathbf{N}.$$

The power series $\sum_0^\infty \binom{\alpha}{k} z^k$ is termed the *binomial series associated with parameter α* . Note that $\binom{\alpha}{k}$ is the usual binomial coefficient of elementary algebra for $\alpha \in \mathbf{N}$. We have an example of formal extension.

Show by considering

$$[S_\alpha \exp \circ (-\alpha\lambda)]'(z), \quad |z| < 1,$$

that

$$S_\alpha(z) = \exp[\alpha \lambda(z)], \quad |z| < 1.$$

In particular, when $\alpha = n^{-1}$, n a positive integer, we have $[S_{1/n}(z)]^n = 1 + z$, that is, $S_{1/n}$ furnishes in power series form an *analytic n th root* of the function $1 + z$ with domain $\Delta(0; 1)$. We shall see in the next section in what sense $S_\alpha(z)$ is to be interpreted as an α th power of $(1 + z)$, $|z| < 1$.

Show, arguing from the differential equation (3.7), that $S_{\alpha+\beta}(z) = S_\alpha(z) S_\beta(z)$, $|z| < 1$. Give a second proof by first establishing the identity

$$\binom{\alpha + \beta}{n} = \sum_0^n \binom{\alpha}{k} \binom{\beta}{n-k}, \quad \alpha, \beta \in \mathbf{C}, \quad n \in \mathbf{N},$$

using the elementary binomial theorem for $\alpha, \beta \in \mathbf{N}$ and properties of polynomials and thereupon Cauchy multiplication.

3.10. $S_{1/2}$. The binomial series, $\alpha = \frac{1}{2}$, is particularly interesting. It plays a useful role in approximation theory in the real domain. Show that $S_{1/2}(x) = \sqrt{1+x}$, $-1 < x < 1$. From $S_{1/2}(-x) > 0$, $0 < x < 1$, conclude that $\sum_0^\infty \left| \binom{1/2}{k} \right|$ is convergent and hence that $\sum_0^\infty \binom{1/2}{k} z^k$ converges uniformly and absolutely on $\overline{\Delta(0; 1)}$. The sum on $[-1, 1]$ is $\sqrt{1+x}$. Show, noting that $-1 \leq x^2 - 1 \leq 0$ for $-1 \leq x \leq 1$, that $|x|$ may be approximated uniformly on $[-1, 1]$ by a polynomial.

3.11. Let $f(z) = \sum_1^\infty a_k z^k$, $|z| < r$, where $r > 0$. Let $F(z) = \sum_0^\infty |a_{k+1}| z^k$ and let $M = \lim_{r \rightarrow 0^+} F|]0, r[$. It is supposed that $M < +\infty$. Let $g: \Delta(0; s) \rightarrow \Delta(0; r)$, $s > 0$, and satisfy $f \circ g(z) = z$, $|z| < s$. Show that $|g(z)| \geq |z|/M$. Apply to the following situation: n a positive integer, $r = 2^{1/n} - 1$, $s = 1$, $f(z) = (1+z)^n - 1$, $g(z) = S_{1/n}(z) - 1$.

3.12. Seek a power series with radius of convergence $\rho > 0$ whose sum w on $\Delta(0; \rho)$ satisfies $w(0) = 1$ and $z w''(z) + w'(z) + z w(z) = 0$, $|z| < \rho$. Show that there is exactly one such series and $\rho = +\infty$. The differential equation is *Bessel's equation of order 0*, the solution in question, customarily denoted J_0 , is the *Bessel function of order 0*.

Carry out the same program when w is to satisfy $w(0) = 1$ and

$$z(1-z)w''(z) + [c - (a+b+1)z]w'(z) - abw(z) = 0$$

where $a, b, c \in \mathbb{C}$, $-c \notin \mathbb{N}$ (*hypergeometric equation of Gauss*). Show that $\rho \geq 1$ and that the k th coefficient of the power series in question is

$$\frac{\prod_{j=0}^{k-1} (a+j) \prod_{j=0}^{k-1} (b+j)}{k! \prod_{j=0}^{k-1} (c+j)}.$$

Show that if $b = c = 1$, then $w(z) = S_{-1}(-z)$.

In this exercise the power series approach to linear differential equations is treated *ad hoc*. In Ch. XII we shall see how the majorant method of Cauchy permits us to put the power series approach to differential equations on a systematic basis.

4. FURTHER PROPERTIES AND APPLICATIONS OF THE EXPONENTIAL FUNCTION, TRIGONOMETRIC FUNCTIONS

In this section we treat mapping properties of the exponential function. Further we introduce the sine and cosine arithmetically, that is, in terms of analysis without appeal to geometric considerations. In addition, the notions of logarithm, argument, root, and power will be studied. The latter concepts are elementary, but in practice care is required for their correct use.

From the power series representation of the exponential function we

see that $\exp(\bar{a}) = \overline{\exp(a)}$ and consequently $|\exp(a)| = \exp(\operatorname{Re} a)$. Hence $|\exp(ia)| = 1$ if and only if $a \in \mathbf{R}$. We define *sine* and *cosine* on $\mathbf{C}$ arithmetically by

$$\cos z = (e^{iz} + e^{-iz})/2, \quad \sin z = (e^{iz} - e^{-iz})/2i. \quad (4.1)$$

Hence $\exp(iz) = \cos z + i \sin z$ and $\exp(-iz) = \cos z - i \sin z$, the former equality being the formula of Euler. The function *cis* with domain $\mathbf{C}$ defined by $\operatorname{cis}(z) = \exp(iz)$ is convenient.† By virtue of the addition theorem for the exponential function we have the de Moivre formula

$$\operatorname{cis}(a + b) = (\operatorname{cis} a)(\operatorname{cis} b), \quad a, b \in \mathbf{C}.$$

Taken together with the formula of Euler it yields the standard addition theorems for $\cos(a + b)$ and $\sin(a + b)$.

We now consider the action of *cis* | $\mathbf{R}$. We have from (4.1) $\cos' = -\sin$, $\sin' = \cos$. Consider $\cos(3)$. We have

$$\cos z = \sum_0^{\infty} \frac{(i)^k + (-i)^k}{2(k!)} z^k = \sum_0^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}.$$

Hence

$$\begin{aligned} \cos(3) &= 1 - \sum_0^{\infty} \frac{9^{2n+1}}{(4n+2)!} \left[1 - \frac{9}{(4n+3)(4n+4)} \right] \\ &< 1 - \left(\frac{9}{2}\right)\left(\frac{1}{4}\right) < 0. \end{aligned}$$

Here the estimate is made by considering the zeroth partial sum. On the other hand $\cos(0) = 1$. Hence $\cos x = 0$ for some $x \in]0, 3[$. Let $\pi/2$ be the smallest positive root of $\cos x = 0$. From $\cos x > 0$, $0 < x < \pi/2$, it follows that $\sin$ is increasing on $[0, \pi/2]$ and hence $\sin x > 0$, $0 < x \leq \pi/2$, and $\sin(\pi/2) = 1$.‡ From $\cos' = -\sin$, we conclude that $\cos$ is decreasing on $[0, \pi/2]$. We see that *cis* is *univalent* on $[0, \pi/2]$. Note that $\operatorname{cis}(\pi) = [\operatorname{cis}(\pi/2)]^2 = -1$ and that $\operatorname{cis}(2\pi) = 1$. On using the identity $(\operatorname{cis} t)[\operatorname{cis}(\pi - t)] = -1$, we see that $\cos$ is decreasing on $[\pi/2, \pi]$. Consequently *cis* is univalent on $[0, \pi]$. Further the image of $[0, \pi]$ with respect to *cis* is $C(0; 1) \cap \{\operatorname{Im} z \geq 0\}$. From $(\operatorname{cis} t)[\operatorname{cis}(2\pi - t)] = 1$, we see that *cis* maps $[\pi, 2\pi]$ univalently onto $C(0; 1) \cap \{\operatorname{Im} z \leq 0\}$ and hence that *cis* maps $[0, 2\pi[$ univalently onto $C(0; 1)$.

It is now easy to see that $e^z = 1$ if and only if z is of the form $2\pi in$, $n \in \mathbf{Z}$. For

† “*cis*” = “ $\cos + i \sin$.”

‡ Gloss: when we say that a function f has a certain property on a subset E of its domain, we mean that $f|E$ has the stated property—as “ f increases on E ” or “ f is univalent on E .”

if $e^{it} = 1$, then $t \in \mathbf{R}$, and if m is the maximum of the integers not exceeding $t/2\pi$, we have $2\pi m \leq t < 2\pi(m+1)$. From this inequality and $\text{cis}(t - 2\pi m) = 1$ we see that $t = 2\pi m$. The converse is obvious. Hence we conclude that $e^a = e^b$ if and only if $b = a + 2\pi in$ for some $n \in \mathbf{Z}$.

The image of the exponential function. We know that 0 is omitted. Suppose that $a \neq 0$. Then $a/|a| = \text{cis } \alpha$ for some α , $0 \leq \alpha < 2\pi$. Hence

$$a = \exp(\log |a| + \alpha i).$$

The preimage of $\{a\}$ is $\{\log |a| + (\alpha + 2\pi n)i : n \in \mathbf{Z}\}$. We see that $\exp(\mathbf{C}) = \mathbf{C} - \{0\}$.

EXERCISES

4.1. $\exp$ maps $\{0 \leq \text{Im } z < 2\pi\}$ univalently onto $\mathbf{C} - \{0\}$. The image of $\{\text{Im } z = a\}$ with respect to $\exp$ is a ray less the point 0 and the image of $\{\text{Re } z = a\}$ is $C(0; e^a)$, $a \in \mathbf{R}$. What are the images with respect to $\exp$ of $\{0 \leq \text{Im } z \leq \pi/2\}$, of $\{a < \text{Re } z < b, 0 < \text{Im } z < \pi\}$, $-\infty \leq a < b \leq +\infty$?

4.2. Given $\alpha_1, \dots, \alpha_n$ distinct in $\mathbf{C}$, n a positive integer. Show that if

$$\sum_{k=1}^n P_k(z) \exp(\alpha_k z) = 0, \quad z \in \mathbf{C},$$

where the P_k are polynomials, then the P_k are identically zero. Hint: Take j so that $|\alpha_j| \geq |\alpha_k|$, $k = 1, \dots, n$, and consider the behavior on the ray $\{\tilde{\alpha}_j t : t \in \mathbf{R}^+\}$.

Logarithm, argument. Given $a \in \mathbf{C} - \{0\}$, the solutions of $e^z = a$ are termed the *logarithms* of a , and the solutions of $\text{cis } \varphi = a/|a|$ are termed the *arguments* of a . For a positive $\log a$ of elementary analysis is one of the logarithms of a in the sense just put down. Returning to a not necessarily positive we term the unique logarithm of a whose imaginary part lies in $]-\pi, \pi]$ the *principal logarithm* of a and denote it $\text{Log } a$. For a positive $\log a = \text{Log } a$. We term the unique argument of a in $]-\pi, \pi]$ the *principal argument* of a and denote it $\text{Arg } a$. Clearly $\text{Log } a = \log |a| + i \text{Arg } a$. Arg denotes the function assigning $\text{Arg } z$ and Log the function assigning $\text{Log } z$ to $z \in \mathbf{C} - \{0\}$.

We shall have frequent occasion to associate with a complex-valued function not taking the value 0 continuous or analytic functions whose value at each point of the domain of the given function is a logarithm of the value of the given function, the associated functions being continuous or analytic according to the circumstances. We shall also be concerned with “continuous arguments” in a similar way. Let us be specific. Let $f: X \rightarrow \mathbf{C} - \{0\}$, where X is a topological space and f is continuous on X . Then $g: X \rightarrow \mathbf{C}$ is termed a

continuous logarithm of f provided that g is continuous on X and $f = \exp \circ g$; $h: X \rightarrow \mathbf{R}$ is termed a *continuous argument* of f provided that h is continuous on X and $f/|f| = \text{cis} \circ h$. When X is an open set $O \subset \mathbf{C}$ and f is analytic on O , $g: O \rightarrow \mathbf{C}$ is termed an *analytic logarithm* of f provided that g is analytic on O and $f = \exp \circ g$.

EXERCISES

4.3. g is a continuous logarithm of f if and only if $\text{Re } g = \log |f|$ and $\text{Im } g$ is a continuous argument of f . It is supposed that f is as above.

4.4. If X is connected and g and h are continuous logarithms of f , then there exists $n \in \mathbf{Z}$ such that $h(x) - g(x) = 2\pi in$, $x \in X$. Granting the existence of a continuous logarithm of f , say g , show that the totality of continuous logarithms of f is $\{g + 2\pi in\}$. Here $2\pi in$ is construed as the constant on X taking the value $2\pi in$, $n \in \mathbf{Z}$.

4.5. If f is analytic on O open $\subset \mathbf{C}$ and g is a continuous logarithm of f , then g is an analytic logarithm of f .

Continuity properties of Arg. We shall show: $\text{Arg} \mid \Omega$, where

$$\Omega = \mathbf{C} - \{-t: t \in \mathbf{R}^+\},$$

is continuous. To see this, we note that if O is open in the sense of the relative topology of $]-\pi, \pi[$, then $\text{cis}(O) = C(0; 1) - \text{cis}(E)$, where E is the compact set $[-\pi, \pi] - O$. Hence $\text{cis}(O)$ is open in the sense of the relative topology of $C(0; 1) - \{-1\}$. Hence Arg restricted to $C(0; 1) - \{-1\}$ is continuous. Since $\text{Arg } z = \text{Arg}(z/|z|)$, $z \neq 0$, we conclude our assertion. The continuity of $\text{Log} \mid \Omega$ follows.

We go a step further: $\text{Log} \mid \Omega$ is analytic and is a primitive of $1/z$ with domain Ω . To that end, let $a \in \Omega$ and let $b = \text{Log } a$. Using the Carathéodory formulation of the derivative we have

$$e^z - e^b = \varphi(z)(z - b),$$

where φ is analytic on $\mathbf{C}$ and $\varphi(b) = e^b \neq 0$. From

$$z - a = \exp(\text{Log } z) - \exp(b) = \varphi(\text{Log } z)(\text{Log } z - \text{Log } a),$$

we see that $\text{Log}'(a) = 1/\varphi(\text{Log } a) = 1/a$. We shall meet this argument again in the study of the inverse of an analytic function (cf. Ex. 9.6, Ch. V).

EXERCISES

4.6. Show that λ of §3, this chapter, satisfies $\lambda(z - 1) = \text{Log } z$, $|z - 1| < 1$.

4.7. There exists a unique analytic logarithm of the identity on $\{\text{Re } z < 0\}$

which agrees with Log on $\{\text{Re } z < 0\} \cap \{\text{Im } z > 0\}$. It differs from Log on $\{\text{Re } z < 0\} \cap \{\text{Im } z < 0\}$ by the constant value $2\pi i$. Hint: $\text{Log}(-z) + \pi i$.

4.8. Given $a \in \mathbb{C} - \{0\}$. Let α be a logarithm of a . Then $\text{Log}(z/a) + \alpha$ is an analytic logarithm of the identity on $\mathbb{C} - \{-at: t \in \mathbb{R}^+\}$.

Polar representation. By the *polar representation* of $\mathbb{C}$ we mean the map of $\mathbb{R}^+ \times \mathbb{R}$ into $\mathbb{C}$ taking (r, θ) into $re^{i\theta}$. Its image is $\mathbb{C}$. Indeed, the preimage of $\{0\}$ is $\{0\} \times \mathbb{R}$ and the preimage of $\{z\}$, $z \neq 0$, consists of the set of ordered pairs $(|z|, \alpha)$ where α is an argument of z .

Roots of a complex number. Let $a \in \mathbb{C}$; let n be a positive whole number. We ask for the roots of the equation

$$z^n = a. \quad (4.2)$$

They are termed *the n th roots of a* . If $a = 0$, the situation is simple. There is precisely one solution. It is 0. If $a \neq 0$ and z is a solution so that $z \neq 0$, setting $z = |z|e^{i\theta}$, by (4.2) we see that $|z| = \sqrt[n]{|a|}$ and $n\theta$ is an argument of a . Hence θ must be of the form

$$\frac{1}{n} \text{Arg } a + 2\pi \frac{k}{n}, \quad k \in \mathbb{Z}.$$

The solutions of (4.2) are numbers of the form

$$\sqrt[n]{|a|} \text{cis}\left(\frac{\text{Arg } a}{n}\right) \text{cis}\left(\frac{2\pi k}{n}\right), \quad k \in \mathbb{Z}. \quad (4.3)$$

Conversely, every number of the form (4.3) satisfies (4.2). Actually, there are precisely n elements of the form (4.3), namely, those obtained by considering $k \in \langle 0, n-1 \rangle$. Special case: $a = 1$. Here we obtain *the n th roots of unity*. We see that if $\omega = \text{cis}(2\pi/n)$, then the n th roots of unity are given by ω^k , $k \in \langle 0, n-1 \rangle$. The n th roots of unity form a cyclic subgroup of the multiplicative group $\mathbb{C} - \{0\}$. If b is any n th root of a , then the totality of n th roots of a is given by $b\omega^k$, $k \in \langle 0, n-1 \rangle$.

EXERCISES

4.9. Determine ω explicitly for $n = 3, 4, 5, 6, 8$.

4.10. Show that if n is prime, then for each $k \in \langle 1, n-1 \rangle$ the cyclic group of the n th roots of unity is generated by ω^k .

4.11. Show that the continuous n th roots of $1+z$ with domain $\Delta(0; 1)$ are given by $\omega^k S_{1/n}$, where $\omega = \text{cis}(2\pi/n)$. Show that the continuous n th roots of the identity map on $\Omega = \mathbb{C} - \{-t: t \in \mathbb{R}^+\}$ are given by $\omega^k \exp \circ [(1/n) \text{Log}] \Omega$.

4.12. Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$, $|z| < r$, where $r > 0$, $a_0 = 1$, and $\{k: a_k \neq 0, k > 0\}$ is not

empty. Let m denote the minimum positive k satisfying $a_k \neq 0$. Show that with $|u| = 1$ we have

$$\lim_{t \rightarrow 0} [|f(tu)|^2 - 1] t^{-m} = 2 \operatorname{Re}(a_m u^m).$$

Here t is assumed to take small positive values. Conclude using the arbitrariness of u that $|f|$ possesses neither a relative maximum nor a relative minimum at 0.

Apply to furnish a proof of the Fundamental Theorem of Algebra by showing that if P is a polynomial of positive degree and $P(a) \neq 0$, then $|P|$ does not have a relative minimum at a . This beautiful piece of 18th century mathematics is due to d'Alembert.

Powers. Once more we are faced, as we were in the study of the exponential function, with a problem of "reasonable" definition. What is to be meant by a complex power of a complex number? "Reasonable" means that we want to have certain requirements fulfilled. These include: (1) normalization, as we know it with exponents in $\mathbf{N}$, $a^0 = 1$, $a^1 = a$; (2) the usual addition theorem. While the powers 0^n , $n \in \mathbf{N}$, are convenient, for $a = 0$ we lose multiplicative meaning when we consider as candidate exponents the negative integers. In what follows we put the case $a = 0$ completely aside.

We fix $a \in \mathbf{C} - \{0\}$ and ask for f analytic on $\mathbf{C}$ satisfying: (1) $f(z + w) = f(z)f(w)$, $z, w \in \mathbf{C}$; (2) $f(1) = a$. Given $b \in \mathbf{C}$, by the b th powers of a are meant the values $f(b)$, all admitted f being taken into account. Our task is to determine the f in question.† We see that f fulfilling the stated conditions does not take the value 0 anywhere. Further by (1) we have

$$f(z + w) - f(z) = f(z) [f(w) - f(0)],$$

whence

$$f'(z) = c f(z), \quad z \in \mathbf{C},$$

where $c = f'(0)$. Now the derivative of $f(z) \exp(-cz)$ is identically zero on $\mathbf{C}$. Hence $f(z) \exp(-cz) = f(0) = 1$ and $f(z) = \exp(cz)$. Setting $z = 1$, we see that c is a logarithm of a . Conversely, if c is a logarithm of a , the function $\exp(cz)$ with domain $\mathbf{C}$ fulfills the stated conditions. Thus given $a \in \mathbf{C} - \{0\}$, $b \in \mathbf{C}$, the b th powers of a are the numbers

$$\exp[b(\operatorname{Log} |a| + i \operatorname{Arg} a)] \exp(2\pi i k b), \quad k \in \mathbf{Z}.$$

We see that the n th roots of $a \in \mathbf{C} - \{0\}$ are the $(1/n)$ th powers of a , n a positive integer. Also we note that with $a = e$ the exponential function is an

† The stipulation "analytic" may appear severe. In Ex. 5.9, Ch. V, the question of replacing "analytic" by "continuous" will be considered. If we merely ask for $f: \mathbf{C} \rightarrow \mathbf{C}$ satisfying (1) and (2) but no supplementary condition, we encounter solutions of a highly discontinuous sort related to the discontinuous solutions of $f(x + y) = f(x) + f(y)$ given by Hamel [42, p. 150].

allowed f . Hence the traditional e^z for $\exp(z)$ is defensible. We reserve e^z systematically to mean $\exp(z)$ (which is one of the z th powers of e).

Continuous and analytic powers. Given $f, g: X \rightarrow \mathbb{C}$, where X is a topological space, f and g are continuous on X , and f does not take the value 0 anywhere, we say that $h: X \rightarrow \mathbb{C}$ is a *continuous g th power of f* provided that h is continuous on X and that $h(x)$ is a $g(x)$ th power of $f(x)$ for each $x \in X$. When g is constant, we allow the obvious licenses of notation. In the special case where X is an open set $O \subset \mathbb{C}$ and f and g are analytic on O , h is called an *analytic g th power of f* provided that h is analytic on O and $h(x)$ is a $g(x)$ th power of $f(x)$ for each $x \in O$.

When we speak of continuous (analytic) n th roots, the definitions follow the above with “ n th root” replacing “ g th” (or “ $g(x)$ th”) “power” save that we do not require that f take only values that are not zero.

EXERCISES

4.13. Determine the b th powers of a for the following cases of (a, b) : $(1, i\alpha)$ where $\alpha \in \mathbb{R}$, $(e, \pi i)$, (i, i) , (e, z) .

4.14. Show that S_α is an analytic α th power of the function $1 + z$ with domain $\Delta(0; 1)$. *[Determine all the continuous α th powers of $1 + z$ with domain $\Delta(0; 1)$.]

4.15. $\exp \circ (\alpha \operatorname{Log})|_\Omega$ is an analytic α th power of the identity map on $\Omega = \mathbb{C} - \{-t: t \in \mathbb{R}^+\}$. *[Determine all the continuous α th powers of the identity map on Ω .]

5. FOURIER ASPECTS OF POWER SERIES

Given $F: \mathbb{R} \rightarrow \mathbb{C}$ where F is continuous on $\mathbb{R}$ and satisfies the periodicity condition $F(\theta + 2\pi) = F(\theta)$, $\theta \in \mathbb{R}$, by the k th complex Fourier coefficient of F , denoted $C_k(F)$, is meant

$$(2\pi)^{-1} \int_0^{2\pi} F(\theta) e^{-ki\theta} d\theta.$$

Here $k \in \mathbb{Z}$. The relation between F and the sequence $(S_n)_{n \in \mathbb{N}}$ where

$$S_n(\theta) = \sum_{-n}^n C_k(F) e^{ki\theta}$$

is a basic question in the theory of Fourier series. In the present section we consider Fourier series of F whose genesis is of a very special kind. Let $\sum_0^\infty a_k z^k$ have radius of convergence $\rho > 0$ and let f denote its sum on $\Delta(0; \rho)$.

Let $0 \leq r < \rho$. We introduce $F(\theta) = f(re^{i\theta})$. From the fact that

$$(2\pi)^{-1} \int_0^{2\pi} e^{ni\theta} d\theta$$

is 1 when $n = 0$ and is 0 when $n \in \mathbf{Z} - \{0\}$ and the uniform convergence of $\sum_0^\infty a_k z^k$ on $C(0; r)$, we have

$$\begin{aligned} C_k(F) &= a_k r^k, & k \in \mathbf{N}; \\ C_k(F) &= 0, & k \in \mathbf{Z} - \mathbf{N}. \end{aligned}$$

From the fact that S_n tends uniformly to F on $\mathbf{R}$, we obtain

$$\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \lim_{n \rightarrow \infty} \int_0^{2\pi} |S_n(\theta)|^2 d\theta.$$

Since

$$\int_0^{2\pi} |S_n(\theta)|^2 d\theta = 2\pi \sum_0^n |a_k|^2 r^{2k},$$

as we see on expanding $|S_n(\theta)|^2$ and thereupon integrating on $[0, 2\pi]$, we conclude

$$\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = 2\pi \sum_0^\infty |a_k|^2 r^{2k}, \quad (5.1)$$

the Parseval identity for power series. From it we see that, with

$$M(r) = \max_{C(0; r)} |f|,$$

we have

$$|f(0)| = |a_0| \leq \left[(2\pi)^{-1} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta \right]^{1/2} \leq M(r).$$

If $|f(0)| = M(r)$ for some r , $0 < r < \rho$, then $a_{k+1} = 0$, $k \in \mathbf{N}$, and f is constant.

Let us now turn to a function $f: \Delta(0; 1) \rightarrow \mathbf{C}$, continuous on $\overline{\Delta(0; 1)}$, such that $f(z) = \sum_0^\infty a_k z^k$, $|z| < 1$. Nothing is said about the convergence of $\sum_0^\infty a_k z^k$ on $C(0; 1)$. We seek to represent $f(z)$, $|z| < 1$, in terms of $f|C(0; 1)$. From the uniform continuity of f on $\overline{\Delta(0; 1)}$, we see that

$$\int_0^{2\pi} f(e^{i\theta}) e^{-ki\theta} d\theta = \lim_1 \int_0^{2\pi} f(re^{i\theta}) e^{-ki\theta} d\theta. \quad (5.2)$$

Consequently, the left-hand side of (5.2) is $2\pi a_k$, $k \in \mathbf{N}$. Hence for $|z| < 1$ we have

$$\begin{aligned}
 f(z) &= (2\pi)^{-1} \sum_0^{\infty} \int_0^{2\pi} f(e^{i\theta}) (ze^{-i\theta})^k d\theta \\
 &= (2\pi)^{-1} \int_0^{2\pi} f(e^{i\theta}) \left[\sum_0^{\infty} (ze^{-i\theta})^k \right] d\theta.
 \end{aligned}$$

Consequently, using the geometric series we obtain

$$f(z) = (2\pi)^{-1} \int_0^{2\pi} \frac{f(e^{i\theta})}{1 - ze^{-i\theta}} d\theta, \quad |z| < 1, \quad (5.3)$$

the *Cauchy formula for f in polar form*. Note that (5.3) implies that f is determined by $f|C(0; 1)$; such functions, whose restrictions on $C(0; 1)$ are equal, are themselves equal.

EXERCISES

5.1. Justify the steps leading to (5.3) by appeal to the appropriate theorems concerning Riemann integrals and uniform limits.

5.2. Given that g is a continuous map of $C(0; 1)$ into $\mathbb{C}$. Define

$$h(z) = (2\pi)^{-1} \int_0^{2\pi} \frac{g(e^{i\theta})}{1 - ze^{-i\theta}} d\theta, \quad |z| < 1.$$

Show that $h(z) = \sum_0^{\infty} a_k z^k$, $|z| < 1$, where a_k is the k th Fourier coefficient of $g(e^{i\theta})$.

We continue with f considered before the exercises. From (5.2) we see that the left-hand side of (5.2) is 0 when $k \in \mathbb{Z} - \mathbb{N}$. Consequently,

$$f(0) = (2\pi)^{-1} \int_0^{2\pi} \frac{f(e^{i\theta})}{1 - \bar{z}e^{i\theta}} d\theta, \quad |z| < 1. \quad (5.4)$$

Taking conjugates in (5.4) and adding each side of the resulting equality to the corresponding side of (5.3) we obtain

$$f(z) + \overline{f(0)} = (2\pi)^{-1} \int_0^{2\pi} u(e^{i\theta}) \frac{2}{1 - ze^{-i\theta}} d\theta \quad (5.5)$$

and hence

$$f(z) = (2\pi)^{-1} \int_0^{2\pi} u(e^{i\theta}) \frac{1 + ze^{-i\theta}}{1 - ze^{-i\theta}} d\theta + i v(0), \quad |z| < 1, \quad (5.6)$$

where $u = \operatorname{Re} f$, $v = \operatorname{Im} f$. Here use is made of the observation

$$2 = (1 + ze^{-i\theta}) + (1 - ze^{-i\theta})$$

to reduce (5.5) and the equality

$$u(0) = (2\pi)^{-1} \int_0^{2\pi} u(e^{i\theta}) d\theta,$$

which follows on taking real parts in the formula $f(0) = a_0$ obtained from (5.2), $k = 0$. The formula (5.6) is useful because it represents $f(z)$, $|z| < 1$, in terms of $u \mid C(0; 1)$ and $v(0)$. Taking real parts of both sides of (5.6) we obtain the Poisson integral formula for u :

$$u(z) = (2\pi)^{-1} \int_0^{2\pi} u(e^{i\theta}) \frac{1 - |z|^2}{|e^{i\theta} - z|^2} d\theta, \quad |z| < 1. \quad (5.7)$$

EXERCISES

5.3. Let $0 < r < +\infty$ and suppose now that $f: \overline{\Delta(0; r)} \rightarrow \mathbb{C}$ is continuous on $\overline{\Delta(0; r)}$ and admits a power series representation in $\Delta(0; r)$. Obtain the formulas corresponding to (5.3), (5.6), and (5.7).

5.4. Prove the following lemma of Nehari [145]: Let $c_k \in \mathbb{C}$, $\alpha_k \in \mathbb{Z}$, $k = 1, \dots, n$. Let $\lambda \in \mathbb{R}$, $\lambda \neq \alpha_k$, $k = 1, \dots, n$. Then

$$\int_0^{2\pi} \left| \sum_1^n c_k e^{i\alpha_k \theta} \right| d\theta \geq 2 |\sin \pi \lambda| \left| \sum_1^n \frac{c_k}{\lambda - \alpha_k} \right|. \quad (5.8)$$

Hint:

$$\left| \sum_1^n c_k e^{i\alpha_k \theta} \right| = \left| \sum_1^n c_k e^{i(\alpha_k - \lambda)\theta} \right|, \quad \theta \in \mathbb{R},$$

and

$$\int_0^{2\pi} |g(\theta)| d\theta \geq \left| \int_0^{2\pi} g(\theta) d\theta \right|,$$

for g continuous, complex-valued on $[0, 1]$. * [Apply (5.8) to obtain the Hilbert inequality in finite form: Given $a_k, b_k \in \mathbb{R}$, $k = 1, \dots, n$. Then

$$\sum \frac{a_k b_j}{k+j} \leq \pi \left(\sum_1^n a_k^2 \right)^{1/2} \left(\sum_1^n b_k^2 \right)^{1/2},$$

the sum on the left being taken over (k, j) satisfying $1 \leq k, j \leq n$. Hint: Apply (5.8) with $\lambda = \frac{1}{2}$ and $(\sum_1^n |a_k| e^{ik\theta})(\sum_1^n |b_k| e^{ik\theta})$ taking over the role of $\sum c_k e^{i\alpha_k \theta}$, and the Cauchy-Schwarz-Buniakowsky inequality for integrals

$$\left(\int_A^B gh d\theta \right)^2 \leq \left(\int_A^B g^2 d\theta \right) \left(\int_A^B h^2 d\theta \right),$$

which may be established by exactly the same argument as the Cauchy inequality for sums given in §5, Ch. II. The elementary rules for sums are assumed (cf. Ex. 3.1, Ch. V).]

6. FUNCTIONS OF TWO REAL VARIABLES

We bring together some notions and results concerning functions of two real variables, which constitute a close to irreducible basis for a proper understanding, as far as matters pertaining to analyticity are concerned, of the relation between a function f of a complex variable and its *associate* $f \circ \sigma_1$ where σ_1 is the restriction of σ to the preimage with respect to σ of the domain of f . They are: the notion of *differentiability*, a reasonable criterion for *interchange* of order of differentiation, the notion of *exactness*, and a reasonable criterion for it. It should be said that for certain questions great formal elegance accrues from the use of complex differential coefficients, which will be treated in Ch. XI. However, since this book is addressed to the student and since the role of the complex differential coefficient is put in an advantageous light after a preliminary real analysis examination of the underlying questions, we choose to defer the study of complex differential coefficients. Also, the questions treated here appear in many areas of mathematics and should be learned in their real analysis setting. The reader interested in complex differential coefficients may profitably turn to Ch. XI after §8 of this chapter.

We start by recalling some definitions concerning partial differentiation. Let V be a neighborhood of $(a, b) \in \mathbf{R}^2 = \mathbf{R} \times \mathbf{R}$ in the sense of the product topology on $\mathbf{R}^2$ (which is also that induced by the Euclidean metric with distance $\sqrt{[(y_1 - x_1)^2 + (y_2 - x_2)^2]}$ between (x_1, x_2) and (y_1, y_2)). Let $f: V \rightarrow \mathbf{C}$. We introduce $g = \{(x, f(x, b)): (x, b) \in V\}$ and $h = \{(y, f(a, y)): (a, y) \in V\}$. We say (barbarously, but to accede to long usage) that f possesses a (first) *partial derivative with respect to x at (a, b)* provided that $g'(a)$ exists.† In this case, we term $g'(a)$ the *partial derivative of f with respect to x at (a, b)* and denote it $f_x(a, b)$. Similar definitions are to be in force, “ y ” replacing “ x ” and “ $h'(b)$ ” replacing “ $g'(a)$.” We say that f is *differentiable at (a, b)* provided that there exist $A, B \in \mathbf{C}$ and $\varepsilon: V \rightarrow \mathbf{C}$, $\varepsilon(a, b) = 0$, ε continuous at (a, b) such that

$$\begin{aligned} f(x, y) = & f(a, b) + A(x - a) + B(y - b) \\ & + \varepsilon(x, y) \sqrt{[(x - a)^2 + (y - b)^2]}, \quad (x, y) \in V. \end{aligned} \quad (6.1)$$

† There is also available the better modern index terminology and notation for which see [36]. However it is necessary for the mathematical practitioner to be familiar with the old and deep-rooted usages.

Two immediate consequences of the differentiability of f at (a, b) are: (1) f is continuous at (a, b) , (2) $f_x(a, b) = A$, $f_y(a, b) = B$. Thanks to (2) there is precisely one representation of f of the form (6.1). Taken together (1) and the existence of $f_x(a, b)$ and $f_y(a, b)$ do not guarantee the differentiability of f at (a, b) as the example $f(x, y) = \sqrt{|xy|}$, $(a, b) = (0, 0)$, $V = \mathbf{R}^2$ shows.

EXERCISES

6.1. Verify the last assertion.

6.2. Show that the differentiability of f at (a, b) is equivalent to the existence of $A, B \in \mathbf{C}$ such that

$$\lim_{(a,b)} \frac{f(x, y) - [f(a, b) + A(x - a) + B(y - b)]}{\sqrt{[(x - a)^2 + (y - b)^2]}} = 0, \quad (6.2)$$

and that if A and B satisfy (6.2), then $A = f_x(a, b)$ and $B = f_y(a, b)$.

6.3. f is differentiable at (a, b) if and only if both $\operatorname{Re} f$ and $\operatorname{Im} f$ are.

6.4. Formulate appropriately a notion of differentiability for a function of a single real or complex variable in the manner of this section and show that so defined differentiability is equivalent to the existence of a derivative.

A sufficient condition for differentiability. In the study of the Cauchy-Riemann equation(s), to be discussed in §7 of this chapter, one frequently encounters the introduction of an alien note—a smoothness condition whose sole purpose is to guarantee differentiability. As we shall see, a necessary and sufficient condition for the existence of the derivative of a function f of a complex variable may be stated in terms of its associate $f \circ \sigma_1$ (differentiability and the Cauchy-Riemann condition: $i(f \circ \sigma_1)_x = (f \circ \sigma_1)_y$), without introducing extraneous elements. However, a reasonable sufficient condition for differentiability certainly belongs to the repertory of the practicing analyst. One such is

Let $f: I(h) \rightarrow \mathbf{C}$ where

$$I(h) =]a - h, a + h[\times]b - h, b + h[,$$

h positive. Suppose that $f_x(a, b)$ and $f_y(x, y)$ exist, the latter for all $(x, y) \in I(h)$, and suppose that f_y is continuous at (a, b) . Then f is differentiable at (a, b) .

PROOF: We suppose that f is *real-valued* since we shall want to use the mean-value theorem. The general case follows by taking real and imaginary parts. Given $(x, y) \in I(h)$, using the mean-value theorem, we find

$$\begin{aligned} f(x, y) &= f(a, b) + [f(x, b) - f(a, b)] + [f(x, y) - f(x, b)] \\ &= f(a, b) + [f_x(a, b) + \varepsilon(x)](x - a) + f_y(x, Y)(y - b), \end{aligned}$$

where ε is continuous at a and takes the value 0 there and $Y \in [b, y]$. The quotient on the left-hand side of (6.2) with $A = f_x(a, b)$ and $B = f_y(a, b)$ takes the form

$$\frac{\varepsilon(x)(x-a) + [f_y(x, Y) - f_y(a, b)](y-b)}{\sqrt{[(x-a)^2 + (y-b)^2]}}.$$

We conclude using the condition on ε and the continuity of f_y at (a, b) that f is differentiable at (a, b) . It is to be noted that if $(x, y) \in I(\delta)$, $0 < \delta < h$, then $(x, Y) \in I(\delta)$, so that if, given η positive, δ is so chosen that

$$|f_y(s, t) - f_y(a, b)| < \eta, \quad (s, t) \in I(\delta),$$

then for $(x, y) \in I(\delta) - \{(a, b)\}$ the absolute value of the left-hand side of (6.2) is less than $|\varepsilon(x)| + \eta$. Hence the absolute value of the left-hand side of (6.2) is less than 2η in $I(\delta_1) - \{(a, b)\}$ for some δ_1 , $0 < \delta_1 < \delta$. The assertion follows. In practice, it is common for both f_x and f_y to be continuous on $I(h)$.

EXERCISES

6.5. Given $f: A \rightarrow \mathbb{C}$ where A is a region in $\mathbb{R}^2$ and f_x and f_y both take the value zero at each point of A . Show that f is constant. Hint: Take $I(h) \subset A$ and show that $f|I(h)$ is constant.

6.6. Chain rule. Using the notation of the text consider $f: V \rightarrow \mathbb{C}$ differentiable at (a, b) . Let $\varphi: U \rightarrow V$ where U is an $\mathbb{R}$ -neighborhood of $c \in \mathbb{R}$ and $\varphi(c) = (a, b)$. Let $\varphi(t) = (X(t), Y(t))$. Suppose that both X and Y possess a derivative at c . Then $f \circ \varphi$ possesses a derivative at c and

$$(f \circ \varphi)'(c) = f_x(a, b) X'(c) + f_y(a, b) Y'(c).$$

Suppose now that U is an $\mathbb{R}^2$ -neighborhood of (c, d) , that $\varphi(c, d) = (a, b)$, $\varphi(s, t) = (X(s, t), Y(s, t))$, and that X and Y are differentiable at (c, d) . Then $f \circ \varphi$ is differentiable at (c, d) .

Without any doubt the best approach to differentiable functions is a dimension-free one, such as that in [36]. The present treatment is concerned only with our immediate needs and the setting within which we are operating.

Integrals involving a parameter. Let $a, b, c \in \mathbb{R}$, $a < b$, let h be positive, let $I = [a, b] \times]c - h, c + h[$, and let $f: I \rightarrow \mathbb{C}$, f being continuous on I . Suppose that $f_y(x, y)$ exists for each $(x, y) \in I$ (the definition at points of $\{x = a\} \cap I$ and $\{x = b\} \cap I$ being the same as that given earlier, though, to be sure, I is not a neighborhood of such points). We suppose that f_y is continuous at each point of $[a, b] \times \{c\}$. Now define g by

$$g(y) = \int_a^b f(x, y) dx, \quad c - h < y < c + h. \quad (6.3)$$

We claim that

$$g'(c) = \int_a^b f_y(x, c) dx. \quad (6.4)$$

To see this, we note that, given η positive, there exists δ , $0 < \delta \leq h$, such that

$$|f_y(x, y) - f_y(x, c)| < \eta \quad (6.5)$$

when $|y - c| < \delta$. This is an immediate consequence of the Heine-Borel-Lebesgue theorem applied to $[a, b] \times \{c\}$ which may be covered with squares

$$S(x) =]x - \delta(x), x + \delta(x)[\times]c - \delta(x), c + \delta(x)[$$

such that for $(s, t) \in S(x) \cap I$ we have $|f_y(s, t) - f_y(x, c)| < \eta/2$. Here $\delta(x)$ is taken to be less than h . Suppose that $\{S(x_1), \dots, S(x_n)\}$ is a finite subcovering. It suffices to take $\delta = \min\{\delta(x_1), \dots, \delta(x_n)\}$.

Now note that

$$g(y) - g(c) - (y - c) \int_a^b f_y(x, c) dx = \int_a^b [f(x, y) - f(x, c) - f_y(x, c)(y - c)] dx.$$

By the mean-value theorem applied to the real and imaginary part of $f(x, y) - f(x, c)$ and (6.5) we have

$$\left| g(y) - g(c) - (y - c) \int_a^b f_y(x, c) dx \right| \leq 2^{1/2} \eta (b - a) |y - c|$$

when $|y - c| < \delta$ and (6.4) follows. It is to be observed that (6.4) is applicable to a function whose domain is a subset of $\mathbf{R}^2$ containing I and whose restriction to I fulfills the conditions imposed on f . Similarly, it is easy to see that such theorems hold on one-sided domains such as $[a, b] \times [c, d]$, $c < d$.

EXERCISE

6.7. The following theorem is to be established with the aid of the above rule for differentiating under the integral sign.

Let A be a region in $\mathbf{C}$ and let $f: A \rightarrow \mathbf{C}$ be locally of *power series type*; that is, for each $c \in A$ there exists $(a_k(c)): \mathbf{N} \rightarrow \mathbf{C}$ such that $f(c + z) = \sum_0^\infty a_k(c) z^k$ for z small. Then if $A = \mathbf{C}$, $\sum_0^\infty a_k(c) z^k$ has radius of convergence $+\infty$ and $f(c + z) = \sum_0^\infty a_k(c) z^k$, $z \in \mathbf{C}$, and if $A \neq \mathbf{C}$ then $\sum_0^\infty a_k(c) z^k$ has radius of convergence at least as large as

$$d = \min\{|z - c|: z \in \mathbf{C} - A\}$$

and $f(c + z) = \sum_0^\infty a_k(c) z^k$, $|z| < d$.

Hint: Consider

$$\alpha(r) = (2\pi)^{-1} \int_0^{2\pi} g(c + re^{i\theta}) d\theta, \quad 0 < r < +\infty$$

(or $0 < r < d$ as the case may be), where $g(z) = f(z)(z - c)^{-k}$, $k \in \mathbf{N}$, and show that $\alpha'(r) = 0$ and conclude that $\alpha(r) = a_k(c)$. Hence $|a_k(c)|r^k \leq \max_{c \in (c; r)} |f|$. This yields the statements on radius of convergence. To complete the proof establish and use the following lemma: *If f is of power series type on A , the sets B and C are open, where*

$$B = \{c: f(c + z) = 0, z \text{ small}\},$$

$$C = \{c: f(c + z) \neq 0, z \neq 0 \text{ small}\},$$

and $A = B \cup C$. Note that the sum on $\Delta(0; \rho)$ of a power series with radius of convergence $\rho > 0$ is locally of power series type.

Partial derivatives of the second order. We continue with $f: V \rightarrow \mathbf{C}$, as in the second paragraph of this section. We suppose that $f_x(x, y)$ exists (in the original sense of the definition) for each (x, y) in some neighborhood of (a, b) . With f_x denoting the function assigning to each (x, y) at which it exists $f_x(x, y)$, we are led to introduce, when they exist, $(f_x)_x(a, b)$ and $(f_x)_y(a, b)$ and we write more simply " $f_{xx}(a, b)$ " for the former and " $f_{xy}(a, b)$ " for the latter. In a similar manner we introduce $f_{yx}(a, b)$ and $f_{yy}(a, b)$. The quantities in question are the *partial derivatives of the second order of f at (a, b)* . $f_{xy}(a, b)$ and $f_{yx}(a, b)$ are termed the *mixed second partial derivatives of f at (a, b)* . The functions f_{xx}, f_{xy}, f_{yx} , and f_{yy} are introduced as f_x and f_y were above.

Given O open $\subset \mathbf{R}^2$, we say that $f: O \rightarrow \mathbf{C}$ belongs to the class $C'(O)$ provided that f_x and f_y have domain O and each is continuous on O . It follows from the sufficient condition for differentiability given above that if $f \in C'(O)$, then f is differentiable at each point of O . A function $f: O \rightarrow \mathbf{C}$ will be said to belong to the class $C''(O)$ provided that $f_x, f_y \in C'(O)$.

The mixed second partial derivatives do not always coincide. A well-known example of nonequality is the following:

$$f(0, 0) = 0; \quad f(x, y) = xy \frac{x^2 - y^2}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

Here $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ exist but are distinct.

EXERCISE

6.8. Verify the last assertion calculating $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$.

When $f \in C''(O)$, we have $f_{xy} = f_{yx}$ as a consequence of the following theorem. The C'' condition is the one we meet in practice.

Theorem 6.1: Given $f: I(h) \rightarrow \mathbb{C}$, $I(h)$ as above. Suppose that both f_{xy} and f_{yx} have domain $I(h)$ and that each is continuous at (a, b) . Then $f_{xy}(a, b) = f_{yx}(a, b)$.

PROOF: Again we consider the case where f is real, the unrestricted case following from the consideration of real and imaginary parts. We introduce

$$\varphi(s) = s^{-2}[f(a + s, b + s) - f(a, b + s) - f(a + s, b) + f(a, b)]$$

for $0 < s < h$ and consider its limit at 0. We show $\lim_0 \varphi = f_{xy}(a, b)$ and $\lim_0 \varphi = f_{yx}(a, b)$ and conclude the desired equality. Applying the mean-value theorem to $f(t, b + s) - f(t, b)$, $a \leq t \leq a + s$, we obtain

$$\varphi(s) = s^{-1}[f_x(X, b + s) - f_x(X, b)]$$

for some $X \in]a, a + s[$. We fix such an X and apply the mean-value theorem to $f_x(X, t)$, $b \leq t \leq b + s$, obtaining

$$\varphi(s) = f_{xy}(X, Y)$$

where $Y \in]b, b + s[$. We conclude: $\lim_0 \varphi = f_{xy}(a, b)$. It is to be noted that if $(a + s, b + s) \in I(\delta)$, then $(X, Y) \in I(\delta)$. The proof is completed by paraphrase of the above argument for $f(a + s, t) - f(a, t)$, $b \leq t \leq b + s$.

Exactness. The equality of f_{xy} and f_{yx} will be seen to enter into the important question of *exactness* or *compatibility*, that is, of deciding when $A, B \in C'(O)$ are actually f_x and f_y , respectively, for some $f: O \rightarrow \mathbb{C}$. A necessary condition, the exactness condition, is easily given. From $f_x = A$ and $f_y = B$ we have $f_{xy} = A_y$, $f_{yx} = B_x$, and $f \in C''(O)$. Hence

$$A_y = B_x \tag{6.6}$$

(exactness condition). The case where $A(x, y) = y$, $B(x, y) = 2x$, is one for which (6.6) is not fulfilled. It is not possible on topological grounds to give as simple an answer to the converse problem. However, a local solution is available; that is, the following theorem holds.

Theorem 6.2: Given $A, B \in C'[I(h)]$ satisfying (6.6). Then there exists $f \in C''[I(h)]$ satisfying: $f_x = A$, $f_y = B$.

PROOF: We propose f , the function

$$f(x, y) = \int_a^x A(s, b) ds + \int_b^y B(x, t) dt \tag{6.7}$$

with domain $I(h)$. It is easy to see that $f_y = B$. Since $B \in C'[I(h)]$, we conclude, using differentiation under the integral sign,

$$\begin{aligned}
 f_x(x, y) &= A(x, b) + \int_b^y B_x(x, t) dt \\
 &= A(x, b) + \int_b^y A_y(x, t) dt \\
 &= A(x, y),
 \end{aligned}$$

for all $(x, y) \in I(h)$. The theorem follows.

We shall see that Theorem 6.2 plays an important role in relating harmonic and analytic functions.

EXERCISE

6.9. Show with the aid of the following example that the exactness condition and the C' hypothesis on A and B do not guarantee the existence of a desired f when no restriction is imposed on O . We take the domain to be $\mathbf{R}^2 - \{(0, 0)\}$, $A(x, y) = -y/(x^2 + y^2)$, $B(x, y) = x/(x^2 + y^2)$. Suppose that f existed. Let $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$. Calculate

$$f \circ \gamma(2\pi) - f \circ \gamma(0) = \int_0^{2\pi} (f \circ \gamma)'(t) dt$$

by use of the chain rule and obtain a contradiction.

7. THE CAUCHY-RIEMANN EQUATION(S)

The problem that concerns us here is to relate the existence of the derivative of a function of a complex variable with the properties of its *associate* defined at the beginning of §6 of this chapter. Let $a, b \in \mathbf{R}$, let V denote a $\mathbf{C}$ -neighborhood of $\alpha = a + bi$, let $f: V \rightarrow \mathbf{C}$, and let $F = f \circ \sigma_1$, the associate of f . Suppose that $f'(\alpha)$ exists. Using the Carathéodory formulation for the derivative, we obtain

$$F(x, y) - F(a, b) = [f'(\alpha) + \varepsilon(x, y)][(x - a) + i(y - b)], \quad (7.1)$$

$(x, y) \in \sigma^{-1}(V)$, where ε has domain $\sigma^{-1}(V)$, is continuous at (a, b) , and $\varepsilon(a, b) = 0$. By (6.2) of §6, this chapter and (7.1) we see that F is differentiable at (a, b) . Further, $F_x(a, b) = f'(\alpha)$ and $F_y(a, b) = i f'(\alpha)$. Hence

$$F_y(a, b) = i F_x(a, b). \quad (7.2)$$

This is the *Cauchy-Riemann equation in complex form*. Let $U = \operatorname{Re} F$ and

$V = \text{Im } F$. We have: U and V are differentiable at (a, b) and

$$U_x(a, b) = V_y(a, b), \quad U_y(a, b) = -V_x(a, b). \quad (7.3)$$

These are the Cauchy–Riemann equations. It is immediate that the differentiability of F at (a, b) and (7.2) are equivalent to the differentiability of U and V at (a, b) and (7.3).

We now show that the differentiability of F at (a, b) and (7.2) imply the existence of $f'(\alpha)$. Indeed, using σ_1^{-1} we see that

$$f(z) - f(\alpha) = F_x(a, b) \text{Re}(z - \alpha) + i F_y(a, b) \text{Im}(z - \alpha) + \varepsilon(z) |z - \alpha|,$$

$z \in V$, where ε has domain V and is continuous at α and $\varepsilon(\alpha) = 0$. We conclude that $f'(\alpha) = F_x(a, b)$.

We have

Theorem 7.1: *f possesses a derivative at $\alpha = a + bi$ if and only if F is differentiable at (a, b) and satisfies the Cauchy–Riemann condition (7.2).*

Mention should be made of the sufficient conditions for analyticity given by Montel, Menchoff, and others such as $f: O \rightarrow \mathbb{C}$ be continuous, F_x, F_y exist, and $iF_x = F_y$ on $\sigma^{-1}(O)$, and even somewhat less [82].

EXERCISES

7.1. Show that the Cauchy–Riemann equations and the differentiability condition are satisfied by the following pairs of functions: $x^2 - y^2, 2xy$ on $\mathbb{R}^2$; $e^x \cos y, e^x \sin y$ on $\mathbb{R}^2$; $\log \sqrt{x^2 + y^2}, \text{Arctan}(y/x)$, $x > 0$. Here Arctan is the inverse of $\tan |] - \pi/2, \pi/2[$ and $\tan z = \sin z / \cos z$. What is the analytic function with which the third pair is associated?

7.2. Let A be a region in $\mathbb{C}$ and let f be analytic on A . Show that if $\text{Re } f$ is constant, then so is f . If there exist $a, b \in \mathbb{R}$, $(a, b) \neq (0, 0)$ such that $a \text{Re } f + b \text{Im } f$ is constant, then f is constant. If the function assigning $\overline{f(z)}$ to $z \in A$ is also analytic on A , then f is constant. Conclude that under the assumption of the first sentence if $|f|$ is constant, then f is constant.

7.3. Directional derivative. Given F differentiable at (a, b) and $(u, v) \in \mathbb{R}^2$ such that $u^2 + v^2 = 1$, by the *directional derivative of F at (a, b) in the direction (u, v)* is meant $(F \circ \gamma)'(0)$ where $\gamma(t) = (a + ut, b + vt)$, t small. It will be denoted $F'(a, b; u, v)$. Show that if $f'(\alpha)$ exists, $\alpha = a + bi$, then the associate F satisfies

$$i F'(a, b; u, v) = F'(a, b; -v, u). \quad (7.4)$$

Show that if F is differentiable at (a, b) and (7.4) holds for some (u, v) , then $f'(\alpha)$ exists.

7.4. Show that $f: A \rightarrow \mathbb{C}$, where A is a region in $\mathbb{C}$, may possess a derivative at each point of a line and yet not be analytic at any point. Verify that this is so for $(\text{Im } z)^2$ with domain $\mathbb{C}$ and the line $\mathbb{R}$.

7.5. Show with the aid of $|F|^2$, where F is the associate of f , that if f is analytic on a region A and $|f|$ is constant, then f is constant.

7.6. The Cauchy–Riemann equation in polar form. Suppose F is as in Ex. 7.3 but that now $(a, b) \neq (0, 0)$. Let $(a, b) = (\rho \cos \beta, \rho \sin \beta)$, $\rho > 0$. Let

$$\Phi(r, \theta) = F(r \cos \theta, r \sin \theta)$$

for (r, θ) near (ρ, β) . Then Φ is differentiable at (ρ, β) . Show that

$$\Phi_\rho(\rho, \beta) = F'(a, b; \cos \beta, \sin \beta)$$

and

$$\rho^{-1} \Phi_\theta(\rho, \beta) = F'(a, b; -\sin \beta, \cos \beta)$$

and conclude that, F being the associate of f , if $f'(\alpha)$ exists, then

$$i \Phi_\theta(\rho, \beta) = \rho^{-1} \Phi_\rho(\rho, \beta).$$

This is the *Cauchy–Riemann equation in polar form* associated with f and (ρ, β) , $\rho = |\alpha|$, β an argument of α . State and prove an appropriate converse theorem.

8. HARMONIC FUNCTIONS

There is a very intimate connection between harmonic functions of two real variables and analytic functions of one complex variable. It is the special nature of this relation that the study of each kind of function benefits from it. Let us define the basic terms before proceeding. Let O be an open subset of $\mathbf{R}^2$. Given $u \in C''(O)$, by the *Laplacian* of u , denoted Δu , is meant $u_{xx} + u_{yy}$. Given $u: O \rightarrow \mathbf{R}$, we say that u is *harmonic on O* provided that $u \in C''(O)$ and satisfies the *Laplace equation*

$$\Delta u = 0. \dagger \tag{8.1}$$

We say that a real-valued function u is *harmonic at $(a, b) \in \mathbf{R}^2$* provided that its domain is a neighborhood of (a, b) and that $u|_O$ is harmonic for some open O that is a subset of the domain of u containing (a, b) . Given O , an open subset of $\mathbf{R}^2$, we say that u is *harmonic in O* provided that u is harmonic at each point of O .

A moment's reflection shows that the mere existence of u_{xx} and u_{yy} together with (8.1) would force us to consider too vast a class of functions. Consider, for example, u defined on $\mathbf{R}^2$ by $u(0, 0) = 0$, $u(x, y) = xy/(x^2 + y^2)^2$, $(x, y) \neq (0, 0)$. Here $\Delta u = 0$ (on $\mathbf{R}^2$) but u is *not continuous* at $(0, 0)$.

Connection between analytic functions and harmonic functions. For the present we shall confine our attention to what is in appearance a restrictive class of analytic functions, those for which *the second derivative is continuous*.

† That is, $\Delta u(x, y) = 0$, $(x, y) \in O$.

Actually, as we shall see in §3, Ch. V, analytic functions have derivatives of all orders and even Taylor expansions, so that the restriction is only illusory. We show

Theorem 8.1: *Let O be an open subset of $\mathbb{C}$. Let $f: O \rightarrow \mathbb{C}$ possess a continuous second derivative.† Let F denote the associate of f . Then $\operatorname{Re} F$ and $\operatorname{Im} F$ are harmonic on $\sigma^{-1}(O)$.*

PROOF: From what we have already seen $F_x = f' \circ \sigma_1 = -iF_y$ and, hence,

$$(F_x)_x = -i(F_x)_y = (-iF_y)_x = -i(-iF_y)_y = f'' \circ \sigma_1.$$

Consequently $F \in C''[\sigma^{-1}(O)]$ and $\Delta F = 0$. We see that both $\operatorname{Re} F$ and $\operatorname{Im} F$ are harmonic on $\sigma^{-1}(O)$.

EXERCISE

8.1. Reference is made to the setting of Theorem 8.1. If f does not take the value 0 anywhere, then $\log |F|$ is harmonic. No calculation is to be made. Hint: Use an analytic logarithm of the identity map suitably.

In the opposite direction we have a local converse.

Theorem 8.2: *Given h positive, $(a, b) \in \mathbb{R}^2$. Let U be harmonic on $I =]a - h, a + h[\times]b - h, b + h[$. Then there exists f analytic on $\sigma(I)$ such that $U = \operatorname{Re} F$, F being the associate of f .*

PROOF: $A = -U_y$ and $B = U_x$ are in $C'(I)$ and satisfy the exactness condition (6.6) since U is harmonic on I . Let V be a function with domain I satisfying $V_x = -U_y$, $V_y = U_x$. Then $F = U + iV$ is the associate of a function f analytic on $\sigma(I)$. The theorem follows.

EXERCISES

8.2. Let $U(x, y) = \log(x^2 + y^2)$, $(x, y) \in \mathbb{R}^2 - \{(0, 0)\}$. Show that, although U is harmonic, there exists no f analytic on $\mathbb{C} - \{0\}$ with associate F satisfying $\operatorname{Re} F = U$. Hint: Use Ex. 6.9, of this chapter. By use of an analytic logarithm of the identity map the direct calculation of ΔU may be avoided.

8.3. Under the assumptions of Theorem 8.2 there exists g analytic on $\sigma(I)$ whose associate G satisfies $\log |G| = U$.

† This hypothesis is guaranteed by the analyticity of f .

Polar coordinates. What has been said about the polar representation of $\mathbb{C}$ goes over, of course, for $\mathbb{R}^2$. We are simply dealing with the map p taking (r, θ) into $(r \cos \theta, r \sin \theta)$, $(r, \theta) \in \mathbb{R}^+ \times \mathbb{R}$. Suppose that O is an open subset of $\mathbb{R}^2 - \{(0, 0)\}$ and that $u \in C''(O)$. Let $\Omega = p^{-1}(O)$. We introduce $U = u \circ (p| \Omega)$ and note that $U \in C''(\Omega)$. By the *Laplacian of u in polar form* is meant the following formula for $(\Delta u) \circ (p| \Omega)$ in terms of the derivatives of U :

$$(\Delta u) \circ (p| \Omega) = U_{rr} + r^{-1}U_r + r^{-2}U_{\theta\theta}. \dagger \quad (8.2)$$

EXERCISE

8.4. Verify that $U \in C''(\Omega)$ and that (8.2) holds. Show that $u \in C''(O)$, u real-valued, is harmonic if and only if U satisfies the equation

$$U_{rr} + r^{-1}U_r + r^{-2}U_{\theta\theta} = 0. \quad (8.3)$$

Show that given a real-valued $U \in C''(\Omega)$ satisfying (8.3) and of period 2π in θ , that is, $U(r, \theta + 2\pi) = U(r, \theta)$, $(r, \theta) \in \Omega$, there exists a unique harmonic u with domain O such that $U = u \circ (p| \Omega)$.

The Gauss mean-value theorem. We apply (8.2) to the study of a function harmonic on an annulus. Let $0 \leq r_1 < r_2 \leq +\infty$. Let u be harmonic on

$$A(0, 0; r_1, r_2) = \{(x, y): r_1 < \sqrt{x^2 + y^2} < r_2\}.$$

We introduce $m(r)$, the *mean value* of u on the circumference with center $(0, 0)$ and radius r , $r_1 < r < r_2$, defined by

$$m(r) = (2\pi)^{-1} \int_0^{2\pi} u(r \cos \theta, r \sin \theta) d\theta. \quad (8.4)$$

The function m satisfies a very simple differential equation on $]r_1, r_2[$. In fact, using the Laplacian in polar form, we obtain from

$$(2\pi)^{-1} \int_0^{2\pi} \Delta u(r \cos \theta, r \sin \theta) d\theta = 0$$

the equation

$$m'' + r^{-1}m' = 0 \quad (8.5)$$

when we take into account the fact that

$$\int_0^{2\pi} U_{\theta\theta}(r, \theta) d\theta = 0.$$

$\dagger$ Here r serves to denote the function assigning r to (r, θ) .

From (8.5) we infer $(rm')' = 0$, and hence there exist $c, d \in \mathbf{R}$ such that $m(r) = c \log r + d$, $r_1 < r < r_2$. Actually (c, d) is unique.

Suppose now that we are considering a function u harmonic on an open subset O of $\mathbf{R}^2$ where O contains

$$\{(x, y): (x - a)^2 + (y - b)^2 \leq \rho^2\}.$$

Here ρ is positive. On considering $u(a + x, b + y)$ on $A(0, 0; 0, \rho)$ we see that

$$(2\pi)^{-1} \int_0^{2\pi} u(a + r \cos \theta, b + r \sin \theta) d\theta \quad (8.6)$$

is of the form $c \log r + d$ and that it has limit $u(a, b)$ at 0. Hence $c = 0$. The integral (8.6) is constant, the value being $u(a, b)$. Taking the limit at ρ , we obtain the *Gauss mean-value property* for harmonic functions:

$$u(a, b) = (2\pi)^{-1} \int_0^{2\pi} u(a + \rho \cos \theta, b + \rho \sin \theta) d\theta. \quad (8.7)$$

It is to be emphasized that a, b, ρ are subject to the stated conditions.

The Gauss mean-value property is one means of establishing the important *maximum principle for harmonic functions*: *Let u be harmonic on a region A and let u possess a maximum. Then u is constant.*

PROOF: Let $M = \max_A u$. Let $B = \{u(x, y) = M\}$ and let $C = A - B$. We use the well-known property of the Riemann integral of a continuous real-valued function: *Let $f: [\alpha, \beta] \rightarrow \mathbf{R}^+$ where $\alpha, \beta \in \mathbf{R}$, $\alpha < \beta$, and f is continuous on $[\alpha, \beta]$. If*

$$\int_{\alpha}^{\beta} f dx = 0,$$

then $f(x) = 0$, $\alpha \leq x \leq \beta$. Applied to

$$M - u(a + \rho \cos \theta, b + \rho \sin \theta),$$

$0 \leq \theta \leq 2\pi$, granted

$$\{(x - a)^2 + (y - b)^2 \leq \rho^2\} \subset A$$

and $u(a, b) = M$, it tells us that u takes the value M at each point of $\{(x - a)^2 + (y - b)^2 \leq \rho^2\}$. Since this is so for all sufficiently small positive ρ , we see that B is a neighborhood of (a, b) . Hence B is open. By the continuity of u , $C = \{u(x, y) < M\}$ is open. By the connectedness of A , $B = A$. The maximum principle follows.

Of course, a corresponding minimum principle holds. The two principles have important consequences for harmonic functions which are given in the following exercise.

EXERCISES

8.5. Let A be a region in $\mathbf{R}^2$ and let $u: \bar{A} \rightarrow \mathbf{R}$ be continuous and $u|_A$ be harmonic on A . Here $\bar{A}$ is the closure of A in the sense of the 1-point compactification of $\mathbf{R}^2$; $\text{fr } A$ is to be taken in this sense also. Then $\max_{\bar{A}} u = \max_{\text{fr } A} u$ and $\min_{\bar{A}} u = \min_{\text{fr } A} u$. Show that if v satisfies the above conditions on u and $u|_{\text{fr } A} = v|_{\text{fr } A}$, then $u = v$. Hint: Consider $u - v$. Show with the aid of the example: $A = \{(x, y): 0 < x < +\infty\}$, $u(x, y) = x$, $v(x, y) = 0$, that the theorems of this exercise break down when $\bar{A}$ is taken in the sense of the topology of $\mathbf{R}^2$.

8.6. Let $f(z) = \sum_0^\infty a_k z^k$, $|z| < \rho$, where $\rho > 0$. Suppose that $a_k \neq 0$ for some positive integer k and let m denote the minimum such k . Let $u \in C(0; 1)$. Show that

$$\lim_{r \rightarrow 0} r^{-m} \operatorname{Re}[f(ur) - f(0)] = \operatorname{Re}(a_m u^m).$$

Here we consider $r \in]0, \rho[$. Conclude that $\operatorname{Re} f$ has neither a relative maximum nor a relative minimum at 0.

9. THE THEOREM OF GAUSS ON THE ZEROS OF THE DERIVATIVE OF A POLYNOMIAL

This theorem brings into play the notion of convexity, that of the logarithmic derivative, and the multiplicative decomposition of a polynomial.

Logarithmic derivative. It is worth calling the reader's attention to the fact that this notion will occur many times in the following chapters. Let f possess a derivative at $a \in \mathbf{C}$ and suppose that $f(a) \neq 0$. By the *logarithmic derivative of f at a* is meant $f'(a)/f(a)$. The use of the term "logarithmic" is not bizarre. Indeed, if λ is any analytic logarithm of the identity on $\Delta(f(a); |f(a)|)$ and f is restricted to a sufficiently small neighborhood of a , then the derivative at a of λ composed with f so restricted is precisely $f'(a)/f(a)$. A basic property of the logarithmic derivative is the product property: Given $f, g: V \rightarrow \mathbf{C}$ where V is a neighborhood of $a \in \mathbf{C}$ and f and g possess a derivative at a . Suppose that $f(a)g(a) \neq 0$. Let $h = fg$. Then

$$\frac{h'(a)}{h(a)} = \frac{f'(a)}{f(a)} + \frac{g'(a)}{g(a)}. \quad (9.1)$$

The formula (9.1) extends to finite products. With the obvious appropriate assumptions, when

$$h = \prod_1^n f_k^{\mu_k},$$

where $n \in \mathbf{N}$ and $\mu_k \in \mathbf{Z}$, we have

$$\frac{h'(a)}{h(a)} = \sum_1^n \mu_k \frac{f'_k(a)}{f_k(a)}. \quad (9.2)$$

We leave the multiplicative decomposition of a polynomial to the following exercise.

EXERCISE

9.1. Given a polynomial $P(z) = z^{n+1} + \sum_0^n a_k z^k$, $n \in \mathbf{N}$, $a_k \in \mathbf{C}$. Suppose that c is a zero of P , that is, $P(c) = 0$. Show using the Taylor series of P associated with c that $P(z) = (z - c)(z^n + \sum_0^{n-1} b_k z^k)$ where $b_k \in \mathbf{C}$. Conclude by an inductive argument that there exist $(\mu_k)_1^v$ and $(c_k)_1^v$, v a positive whole number, where each μ_k is a positive integer and $c_k \in \mathbf{C}$ and $c_k \neq c_j$ for $k \neq j$, such that

$$P(z) = \prod_1^v (z - c_k)^{\mu_k}. \quad (9.3)$$

To what extent are v , $(\mu_k)_1^v$, $(c_k)_1^v$ determined? The fundamental equality (9.3) gives the multiplicative decomposition of a polynomial.

We are ready to treat the theorem of Gauss on the zeros of the derivative of a polynomial. It is

Theorem 9.1: *Given P as in Ex. 9.1, the zeros of P' lie in the convex hull of the set of zeros of P .*

PROOF: By (9.2) and the representation (9.3) when $P(z) \neq 0$ we have

$$\frac{P'(z)}{P(z)} = \sum_1^v \frac{\mu_k}{z - c_k} = \sum_1^v \frac{\mu_k}{|z - c_k|^2} (\bar{z} - \bar{c}_k). \quad (9.4)$$

Hence if $P'(z) = 0$ but $P(z) \neq 0$, by (9.4) we conclude, taking conjugates, that

$$z \sum_1^v \frac{\mu_k}{|z - c_k|^2} = \sum_1^v \frac{\mu_k}{|z - c_k|^2} c_k.$$

Hence z is of the form $\sum_1^v \lambda_k c_k$, where $0 < \lambda_k \leq 1$, $k = 1, \dots, v$, and $\sum_1^v \lambda_k = 1$, and consequently z is in the convex hull of $\{c_1, \dots, c_v\}$. If $P'(z) = P(z) = 0$, the assertion is trivial. The proof is complete.

EXERCISE

9.2. Referring to P of Ex. 9.1 and (9.3) show that $a_n = -\sum_1^n \mu_k c_k$ and conclude that the barycenter of the zeros of P' and the barycenter of the zeros of P are equal when $n \geq 1$. Here by the *barycenter* of the zeros of a polynomial of positive degree

$$Q(z) = a \prod_1^n (z - c_k)^{\mu_k},$$

$a \in \mathbb{C} - \{0\}$, is meant $(\sum_1^n \mu_k c_k) / (\sum_1^n \mu_k)$.